

Monitoring using artificial intelligence reveals critical links between housing conditions and respiratory health in pigs

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Abstract Respiratory disease is the most significant issue in the pig industry and leads to high mortality and reduced growth rates. Housing conditions are crucial for maintaining the health and welfare of animals to optimize growth rates. This study aimed to evaluate the respiratory health status of pigs raised in different housing conditions using artificial intelligence (AI) technology and to determine the relationships between respiratory health and environmental factors. Eighty growing pigs (Largewhite × Landrace) × Duroc) were randomly distributed into two housing conditions: the control group, which had normal housing conditions, and the treatment group, which had poor housing conditions. A two-sample t test was performed to compare the groups, and a cross-correlation analysis was conducted to determine associations between respiratory health and days of exposure to environmental factors. The results show significant ($p < 0.001$) lower incidence of respiratory health status along with a significant ($p < 0.001$) greater frequency of coughing in pigs raised under poor housing conditions. Respiratory health is negatively associated with high exposure to temperature, relative humidity, and CO₂ at lags 0 ($R = -0.873, -0.887, \text{ and } -0.869$, respectively), -1 ($R = -0.708, -0.742, \text{ and } -0.705$, respectively), and -2 ($R = -0.545, -0.590, \text{ and } -0.599$, respectively). Furthermore, respiratory health was negatively associated with high exposure to NH₃ ($R = -0.420$) at lag 0. In conclusion, AI technology is an effective tool for monitoring respiratory health. However, further evaluation and calibration of the alarm system threshold are recommended. These findings suggest that maintaining proper control of environmental factors can help prevent respiratory diseases in pigs.

Keywords: smart farming, artificial intelligence, animal welfare, coughing

1. Introduction

Respiratory disease is a paramount concern within the pig industry due to its far-reaching economic implications. This results in substantial financial losses, primarily attributed to increased mortality rates and decreased production performance (Boeters et al 2023). For instance, farrow-to-finish farms in Ireland have experienced profit reductions ranging from 12.8% to 41.0% due to respiratory infections (Calderón Díaz et al 2020). Furthermore, the prevalence of pathogens associated with porcine respiratory diseases remains high in top pork-producing countries, making infections almost inevitable (Arruda et al 2015; Cheong et al 2017; Sun et al 2022). In addition to diminishing performance, respiratory diseases contribute to increased antibiotic usage in both nursery and growing-finishing pigs, driving up medication costs and, consequently, overall production costs (Lekagul et al 2019).

The housing conditions on farms play a pivotal role in enhancing pig productivity and maintaining pig health. Environmental factors, such as temperature, dust, ammonia (NH₃), and carbon dioxide (CO₂), can trigger respiratory diseases in pigs if not managed appropriately (Ruggeri et al 2020). NH₃ has emerged as the most health-threatening pollutant for both animals and humans on farms (Lühken et al 2019). NH₃ acts as a respiratory tract irritant, causing damage and predisposing animals to respiratory infections (Olanrewaju et al 2007). While the standard safety level for NH₃ exposure is set at 25 ppm (Chantziaras et al 2020), research has shown that even at this level, NH₃ can still induce respiratory diseases in pigs (Wang et al 2019). According to Colina et al (2000), maintaining NH₃ concentrations below 10 ppm can effectively prevent health risks in pigs, a goal achievable through efficient ventilation. In general, well-ventilated houses maintain NH₃ levels below 5 ppm (Feng et al 2022; Jo et al 2020). Moreover, maintaining low NH₃ levels can mitigate the risks from other

toxic gases, such as CO₂, carbon monoxide, hydrogen sulfide, and methane (Feng et al 2022; Zong et al 2015).

Regular monitoring of pig health is critical for early disease detection, but conventional methods are costly and invasive and not useful in real time (Lu et al 2020). Timeliness in detection is crucial for preventing further losses due to disease. Therefore, there is a growing effort to develop artificial intelligence (AI) systems for automatic health monitoring based on pig behavior, physiology, or vocalization (Alameer et al 2022; Matthews et al 2017; Shen et al 2022). Infected pigs exhibit elevated temperatures, heart rates, and respiration rates, which can be automatically detected using thermal infrared and RGB cameras (Jorquera-Chavez et al 2020). However, microphone-based AI is more popular than camera-based AI for respiratory disease detection (Lagua et al 2023). In this approach, coughing sounds serve as an indicator of respiratory diseases, with an outstanding accuracy and precision exceeding 95% (Ji et al 2022; Shen et al 2022; Yin et al 2021).

Environmental sensors have been integrated into climate-controlled houses (CCHs) for monitoring and controlling indoor environmental conditions (Van der Stuyf et al., 1989; Yeo et al., 2019). Combining environmental sensors with animal sensors has been proposed as a promising tool for effective disease management and for ensuring the efficiency and sustainability of the swine industry (Pessoa et al 2022). Consequently, the objectives of this study were (1) to assess the respiratory health status of pigs raised in different housing conditions using AI technology and (2) to investigate the associations between respiratory health status and environmental factors such as temperature, relative humidity, NH₃, CO₂, and dust.

2. Materials and Methods

2.1. Animals and Experimental Design

This study received approval from the institutional review board and ethics committee of Suncheon National University, South Korea (SCNU IACUC-2023-20).

The study took place at the Suncheon National University swine experimental farm, where a total of eighty apparently healthy barrows and gilts (large white × Landrace) × Duroc, weighing 28.76 ± 5.09 kg, were used for evaluation. The pigs were assessed for their respiratory health in response to different housing conditions (control and treatment). Two climate-controlled houses equipped with temperature-based mechanical ventilation systems (EURO-500S, Euro Housing Co., Ltd., South Korea) were employed. In the control group, the room temperature was maintained at the ideal temperature for growing pigs, and air conditioning (BUW1450M9S, LG Electronics, South Korea) was used. The slurry pit was regularly emptied every week to prevent the accumulation of toxic gases in the house. In the treatment group, the ventilation rate was reduced, and manure was added to the pit one week before the experiment began to increase the levels of gases such as

NH₃ and CO₂, simulating poor housing conditions. In this group, air conditioning was not used, and the slurry was retained until the end of the experiment.

Each house had four pens with fully slatted plastic floors measuring 2.35×2.90 m, wet-dry feeders (LFS-120, IONTECH Co., Ltd., South Korea) and water troughs. The pigs had unlimited access to feed and water. The houses were illuminated artificially 24 hours a day throughout the 28-day experimental period. Medication for sick pigs was administered only after the experimental period.

2.2. Data Acquisition

2.2.1. Environmental Factor Data

The experimental houses were equipped with environmental sensors placed in the center of each house approximately 2 meters from the floor for the automatic collection of temperature, relative humidity, NH₃, and CO₂ data. The data were accessed online through Farm Note (Narae Trend Co., Ltd. Farm Note, South Korea). However, only CO₂ data from these sensors were utilized. Temperature and relative humidity data were collected using the built-in environmental sensors of the respiratory health monitoring system (SoundTalks NV, Leuven, Belgium). One monitor was installed in each house according to the manufacturer's guidelines. All the collected data were stored and accessed online (<https://www.soundtalksweb.com>). NH₃ concentrations were manually measured using a GASTEC detector tube and pump GV-100 (GASTEC Corporation, Japan). The NH₃ concentration was detected at the pig head level in the center of the house. Additionally, dust particles (PM_{2.5}) were measured using an AR830A SMART SENSOR (Shenzhen Smartsensor Trading Co., Ltd., China).

2.2.2. Respiratory Health Data

A microphone (VideoMicro RODE, Australia) connected to a camera (Hanwha Vision Co., Ltd., South Korea) was installed in each house to record pig sounds during the experiment. The microphones were installed 2 meters from the floor in accordance with the manufacturer's guidelines. The data were saved and accessed online through the Wisenet Recorder Webviewer. From these data, the coughing frequency (CF) was manually counted during the first 5 minutes at 3:00, 6:00, 9:00, 12:00, 15:00, 18:00, 21:00, and 24:00. A total of 40 minutes of audio data per group per day were assessed, starting from the 2nd day until the 27th day. Trained personnel counted the coughs by listening to the recorded audio data.

The respiratory health status (ReHS) of the herd was automatically evaluated using SoundTalks, with one monitor installed in each house. These monitors collected coughing sounds within a 10-meter radius inside the house and sent the data to a gateway wirelessly. Through a sophisticated algorithm, SoundTalks generated a daily ReHS ranging from 0 to 100 (Ko et al 2022). In the previous model, the respiratory distress index (RDI) was generated, representing

the average number of coughs per head within 24 hours (Pessoa et al 2021). The system also included an alarm system to indicate the health conditions of the herd. A ReHS value of at least 60 was considered to indicate a healthy herd. A yellow signal is given for ReHS values between 40 and 59, indicating potential respiratory health problems, and a red signal is given for ReHS values less than 40, indicating a high risk of respiratory health problems.

2.3. Statistical analysis

IBM SPSS Statistics version 20 and RStudio version 2023.06.1 were used for the statistical analyses.

A two-sample *t* test was employed to compare all parameters (temperature, relative humidity, NH₃ concentration, CO₂ concentration, ReHS, and CF) between groups, using days as the experimental units. The significance level (alpha) was set at <0.05. Descriptive statistics were provided for all evaluated parameters.

The respiratory health of pigs is influenced by the duration of exposure to environmental factors. A cross-correlation analysis was performed to examine the association between ReHS and days of exposure (lags) to environmental factors. Only lag 0 and negative lags were considered. Lag 0 corresponded to current-day exposure, while negative lags represented exposure on previous days.

The ReHS value for both groups exceeded 60, indicating no respiratory problems according to the system. However, an augmented Dickey–Fuller (ADF) test was also conducted to determine the stationarity of the daily ReHS data to determine changes in respiratory health. Respiratory health was considered nonstationary or unstable if the *p* value was >0.05. If respiratory health is nonstationary, a sequential Mann–Kendall (SMK) test is performed to identify the change point, which represents the day when a significant change in respiratory health occurred. The inflection point is the intersection point between the prograde and retrograde sequences falling within the 95% confidence intervals.

The CF can be used as a predictor of the ReHS. However, it is unknown whether these two variables have linear or nonlinear relationships. A curve fitting analysis was conducted to determine the relationship between the CF and ReHS. CF was used as the independent variable, and ReHS was the dependent variable.

3. Results

The daily temperature, humidity, NH₃ concentration, CO₂ concentration, and dust concentration are displayed in Figure 1. The pigs in the control and treatment groups were exposed to significantly different ($p < 0.05$) temperatures (24.75 vs 30.05°C), NH₃ concentrations (2.89 vs 20.14 ppm), and CO₂ concentrations (905.92 vs 1,600.93 ppm) (Table 1). However, there were no significant differences ($p > 0.05$) in relative humidity (76.96 vs 79.00%) or dust level (32.00 vs 41.39 mg/m²).

The overall respiratory health of the herds was healthy according to SoundTalks assessments. Yellow and red alarms were not triggered, as both groups had ReHS values exceeding 60, as shown in Table 2. However, the ReHS in the treatment group was significantly lower than that in the control group (86.04 vs. 98.37, respectively; $p < 0.001$). Manual assessment of respiratory health also revealed significant differences ($p < 0.001$) between the groups, with a greater CF observed in the treatment group than in the control group (69.07 vs. 25.07, respectively). Furthermore, the daily ReHS in the treatment group exhibited nonstationary behavior ($p = 0.963$), indicating that the respiratory health of the pigs changed throughout the experimental period. The daily ReHS in the treatment group clearly decreased, as illustrated in Figure 2. According to the sequential Mann–Kendall (SMK) test, the change point for ReHS occurred on the 13th day of the experimental period. The change point for the CF trend in the treatment group coincided with the change point in the ReHS test (Figure 3).

Figure 6 shows the results of the curve fitting analysis of ReHSs and CFs, revealing a strong negative nonlinear relationship (cubic $R^2 = 0.719$).

As depicted in Figure 1, the temperature and the NH₃ and CO₂ concentrations increased in the treatment group. These increases in environmental factors likely contributed to the increased CF and decreased ReHS in the same group (Figure 2). This relationship was confirmed by cross-correlation analysis (Figure 4 and 5). As shown in Figure 5, the temperature, relative humidity, and CO₂ concentration in the treatment group had very strong ($R > -0.800$) to strong ($R > -0.600$) negative associations with ReHS at lags 0 and -1, respectively. A moderate ($R > -0.400$) negative association was observed at lag -2. This finding suggested that the respiratory health of the pigs is influenced by environmental factors from the same day or 1 to 2 days prior. Additionally, the NH₃ concentration was moderately associated ($R = -0.420$) with ReHS at lag 0. However, no significant association was found between ReHS and dust. In contrast, there was no association between ReHS and environmental factors in the control group, except for temperature at lag -11 ($R = 0.421$).

4. Discussion

The objectives of this study were to assess the respiratory health of pigs raised in different housing conditions using a new artificial intelligence (AI) technology model and to determine the associations between respiratory health status and environmental factors, including temperature, relative humidity, NH₃ concentration, and CO₂ concentration.

Overall, the respiratory health of the pigs in both groups was found to be healthy. Nevertheless, the results demonstrated that exposure to elevated levels of environmental factors significantly reduced the respiratory health of the pigs, a phenomenon effectively detected by AI technology. This observation is consistent with previous studies by Pessoa et al. (2022) and Pessoa et al. (2021), in

which they employed the earlier model of SoundTalks in commercial finishing farms and reported increased respiratory distress with increasing environmental factors, specifically NH₃. This technology was also used to monitor the respiratory health of pigs infected with *Mycoplasma hyopneumoniae*, revealing that RDI values change as the

infection progresses (Clavijo et al 2021). Moreover, the highest RDI values corresponded to the highest amount of pathogen detected in the oral fluid samples. These findings underscore the effectiveness of AI technology for detecting respiratory health issues in pigs, whether induced by environmental factors or pathogens.

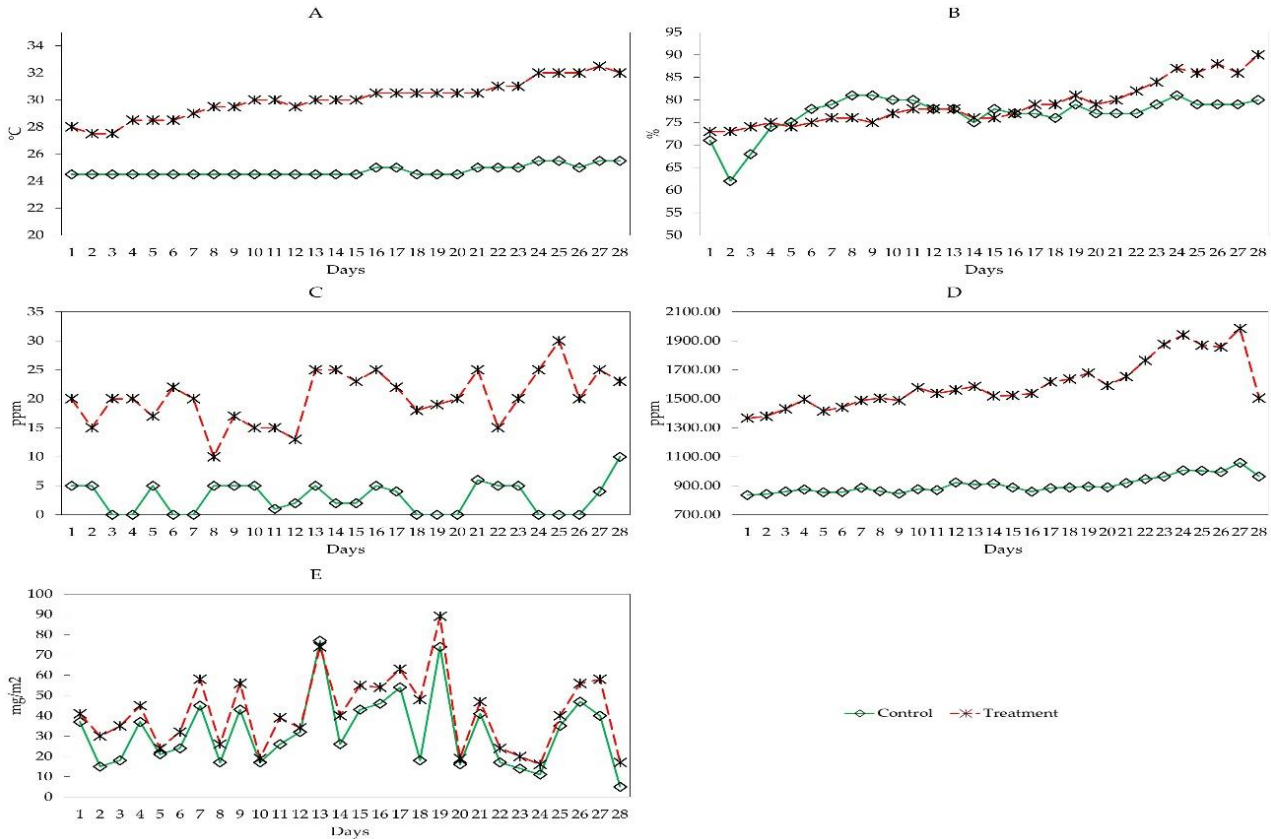


Figure 1 Daily (a) temperature, (b) humidity, (c) ammonia, (d) carbon dioxide, and (e) dust levels in the control and treatment groups.

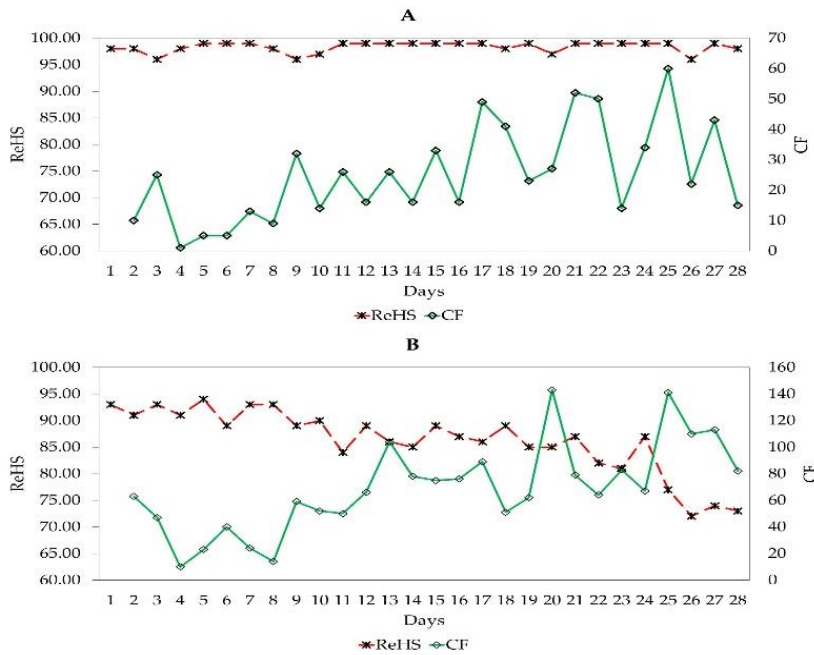


Figure 2 Daily respiratory health status (ReHS) and coughing frequency (CF) of pigs in the (a) control and (b) treatment groups.



Table 1 Differences in the environmental factors between the control and treatment groups.

Parameters		Control	Treatment	SEM	P value
Temperature (°C)	mean ±sd	24.75 ±0.37	30.05 ±1.37	0.381	<0.001
	min–max	24.5–25.50	27.50–32.50		
Humidity (%)	mean ±sd	76.96 ±4.14	79.00 ±4.79	0.609	0.095
	min–max	62.00–81.00	73.00–90.00		
NH ₃ (ppm)	mean ±sd	2.89 ±2.70	20.14 ±4.50	49.790	<0.001
	min–max	0.00–10.00	10.00–30.00		
CO ₂ (ppm)	mean ±sd	905.92 ±57.20	1,600.93 ±170.50	49.790	<0.001
	min–max	836.37–1,058.68	1,366.64–1,985.47		
Dust (mg/m ²)	mean ±sd	32.00 ±17.92	41.39 ±18.33	2.482	0.058
	min–max	50.00–77.00	16.00–89.00		

NH₃= ammonia; CO₂= carbon dioxide.

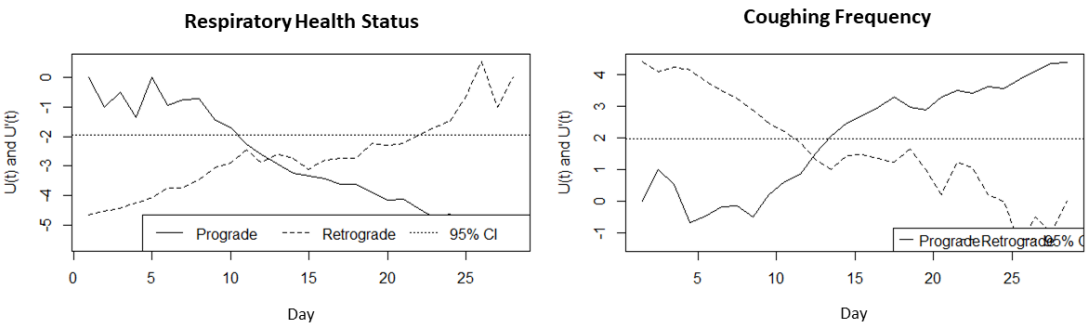


Figure 3 Sequential Mann–Kendall Plot of Respiratory Health Status and Coughing Frequency of Treatment Group. CI: confidence interval. The intersection indicates the change point.

Table 2 Comparison of the respiratory health of growing pigs in the control and treatment groups.

Parameters		Control	Treatment	SEM	P value
ReHS	mean, sd	98.37 ±1.04	86.04 ±5.93	1.023	<0.001
	min–max	96–99	72–94		
	ADF	0.039	0.963		
CF	mean, sd	25.07 ±15.85	69.07 ±33.80	4.668	<0.001
	min–max	1–60	10–143		

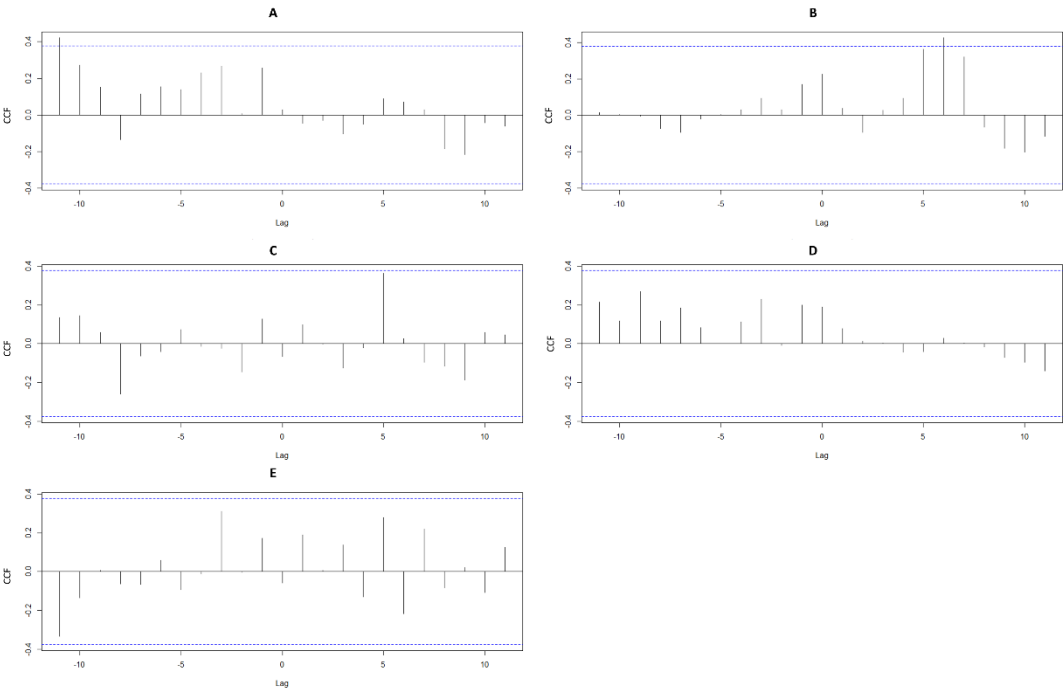


Figure 4 Cross-correlation analysis of respiratory health status and (a) temperature, (b) relative humidity, (c) NH_3 , (d) CO_2 , and (e) dust in the control group. The blue lines indicate significant associations.

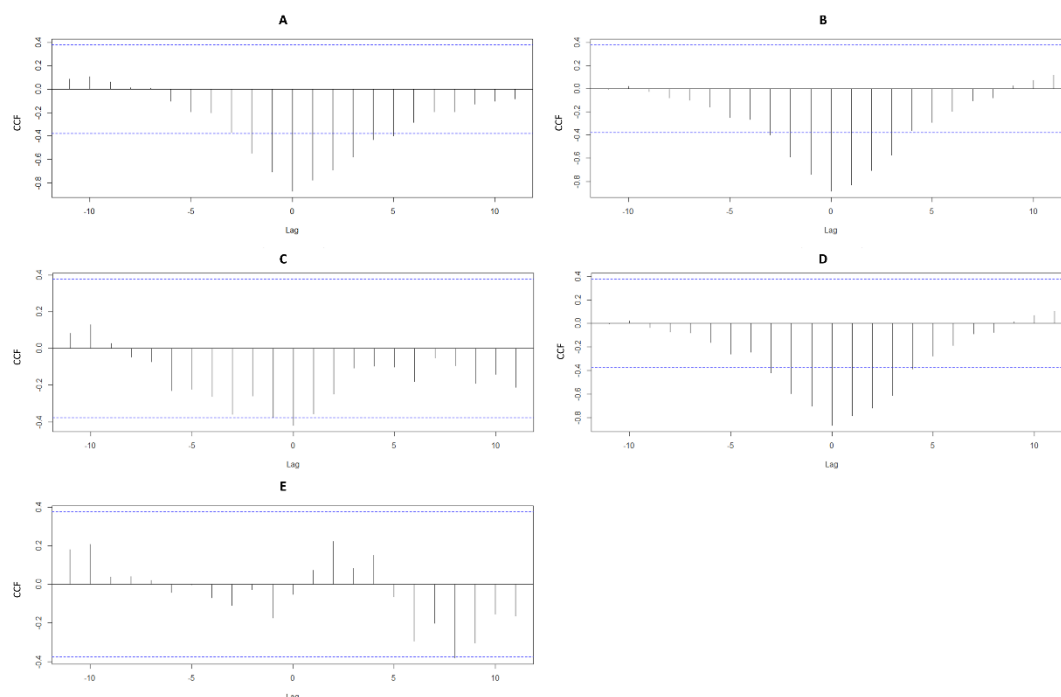


Figure 5 Cross-correlation analysis of respiratory health status and (a) temperature, (b) relative humidity, (c) NH_3 , (d) CO_2 , and (e) dust in the treatment group. The blue lines indicate significant associations.

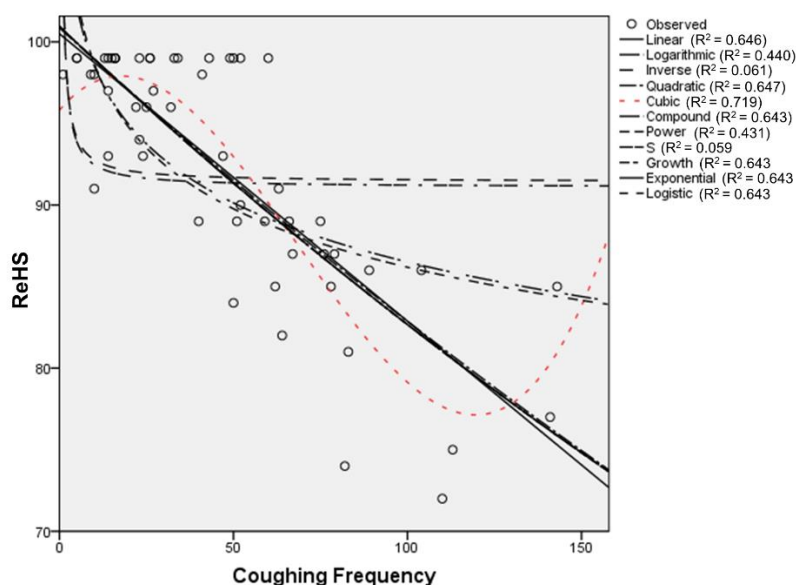


Figure 6 Curve fitting analysis of respiratory health status and coughing frequency.

The sensitivity of the alarm system is crucial for early disease detection using AI. Our study suggested the need for calibration of the threshold of the current technology's alarm system. During the experimental period, no alarms were triggered for potential respiratory problems, as the lowest ReHS recorded was 72. However, based on these trends, the ReHS in the treatment group would likely fall below 60 if the growing period were extended, which could

be too late to notify farmers of a potential respiratory problem. The ReHS in the treatment group exhibited nonstationary behavior, indicating a reduction in respiratory health. The authors explored the potential use of the Sequential Mann–Kendall test in complementing the alarm system for early disease detection, and the test indicated a changepoint on the 13th day. While the coughing frequency

was not continuously collected, the same changepoint was determined based on the SMK test.

Alarm signals were not observed in pigs inoculated with *M. hyopneumoniae*, despite mortality occurring on the 10th day after inoculation (Silva et al 2022). Additionally, the RDI values were lower (ranging from 0.01 to 0.17) than those in other studies, which reported values as high as 23 and 31.87 but reported low levels of coughing during the finisher stage (Pessoa et al 2022; Polson et al 2019). This variation in the generated RDI values and the alarm system threshold may result from differences in the algorithms used in the previous models and from variations in population size. RDI values are room- or farm dependent, as the threshold of the alarm system in the previous model was based on the history and variation of RDI in specific rooms (Pessoa et al 2021). The authors lack information regarding whether the current model uses a similar concept for generating ReHS values. However, the presence of a nonlinear relationship with coughing frequency suggested a similarity (Figure 6). The population size and length of the growing period may also influence the ReHS value and the alarm system. Furthermore, RDI patterns are associated with specific pathogens, and there are different RDI patterns in pigs infected with influenza A virus or *M. hyopneumoniae* (Polson et al 2019). This technology holds potential for more precise disease detection, but there is limited information available on ReHS patterns. Therefore, it is necessary to explore the behavior and patterns of ReHS throughout the production stages in various populations, at different infection levels, for different pathogens, and under diverse environmental conditions.

The cross-correlation analysis demonstrated that respiratory health was negatively affected by temperature, relative humidity, and CO₂ levels at lags 0, -1, and -2 in the treatment group. NH₃ levels were manually assessed, and a moderate negative correlation ($R = -0.420$) with respiratory health was observed at lag 0. In the study by Pessoa et al. (2022), they found associations between respiratory health and NH₃ and relative humidity at negative lags (-1, -7, -8, and -15; and -3, respectively) in one replicate and only with relative humidity at lags -7 and -8 in another replicate. However, no associations were observed in the other replicates. In the present study, temperature, relative humidity, NH₃, and CO₂ levels were consistently and significantly greater in the treatment group than in the control group, except for relative humidity. These findings suggest that exposure to certain levels of environmental factors can induce respiratory problems. The respiratory health of the pigs in the treatment group decreased with increasing incidence of cough (Figure 2). In this group, NH₃ concentrations exceeded 15 ppm (with a maximum of 30 ppm) and showed an increasing trend. Wang et al (2019) also reported an increase in the incidence of cough with NH₃ concentrations up to 25 ppm, which could triple when exposed to 100 and 150 ppm (Stombaugh et al 1969). Although the occupational safety standard for NH₃ is set at 25 ppm for 8 hours of exposure (Chantziaras et al 2020), our

study indicates that even at this level, there can be health risks to pigs. Pigs in the control group were exposed to NH₃ at concentrations less than 10 ppm and exhibited fewer coughing incidents and better respiratory health. These findings align with the statement of Colina et al (2000), who suggested that health risks are prevented at NH₃ levels less than 10 ppm.

Furthermore, the deterioration of respiratory health in the treatment group may be exacerbated by heat stress (resulting from high ambient temperature and high humidity) and high CO₂ levels, as shown in figure 1. Although the experiment was not designed to quantify the effects of each environmental factor, the authors hypothesize that these factors may have indirect effects or synergize with NH₃ to impact the respiratory health of the pigs. Several studies have shown that pigs exposed to chronic heat stress exhibit a depressed immune system, increasing their susceptibility to infection (Cui et al 2016; Huo et al 2019; Ortega & Szabo 2021). In addition, CO₂ concentrations exceeding 1,000 ppm can alter the innate immune responses of the lungs to irritants and potentially exacerbate respiratory problems (Schneberger et al 2021). The CO₂ concentrations recorded in the present study (ranging from 1,366.64 to 1,985.47 ppm) are similar to the slurry pit air CO₂ concentrations recorded by Ni et al (2000), which ranged from 1,228 to 1,556 ppm.

Surprisingly, no association was found between respiratory health and dust concentration. Dust is a common pollutant in swine farms and causes respiratory health problems (McClendon et al 2015). However, dust data were manually collected and not continuously collected in the present study.

5. Conclusions

This study investigated the respiratory health of pigs under different housing conditions without pathogens. AI technology was used for the automatic evaluation of respiratory health alongside manual assessments. The results of the present study demonstrated the efficacy of detecting the respiratory health of pigs using AI technology. However, the results suggest further evaluation and calibration of the threshold of the alarm system. Furthermore, respiratory health is significantly affected by certain threshold levels of environmental factors, such as temperature, relative humidity, NH₃, and CO₂. The findings of the study emphasize the importance of integrating environmental sensors and animal sensors as tools for improved management, which is crucial for maintaining the health and welfare of animals.

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Author's Contribution

Eddiemar B. Lagua and Hong-Seok Mun distributed equally as co-first authors of this manuscript. All authors have read and agreed to the published version of the manuscript.

Ethical considerations

The animal study protocol was approved by the Institutional Review Board or Ethics Committee of Suncheon National University (SCNU IACUC-2023-20).

Conflict of interest

The authors declare no conflicts of interest.

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